

```

timescale 1ns / 1ps
`default_nettype none
`include "parameters.vh"

module imm_gen(
    input wire clk,
    input wire rst_n,
    input wire [31:0] i_Instr,
    output wire [31:0] imm_data
);

    // Extract opcode
    wire [6:0] opcode;
    assign opcode = i_Instr[6:0];

    // Combinational logic for immediate data
    assign imm_data = (opcode == `I || opcode == `LD || opcode == `JR) ?
{{20{i_Instr[31]}}, i_Instr[31:20]} : // I-type
(opcode == `S) ? {{20{i_Instr[31]}}, i_Instr[31:25],
i_Instr[11:7]} : // S-type
(opcode == `B) ? {{19{i_Instr[31]}}, i_Instr[31], i_Instr[7],
i_Instr[30:25], i_Instr[11:8], 1'b0} : // B-type
(opcode == `J) ? {{11{i_Instr[31]}}, i_Instr[31],
i_Instr[19:12], i_Instr[20], i_Instr[30:21], 1'b0} : // J-type
(opcode == `U || opcode == `UPC) ? {i_Instr[31:12], 12'b0} :
// U-type
(opcode == `AMO) ? {{20{1'b0}}, i_Instr[31:27],
i_Instr[24:20]} : // **AMO Immediate (AMO type + offset)**
32'h00000000; // Default for undefined opcodes

endmodule

```